

PROYECTO FINAL DEEP LEARNING

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1. Introduction

This project addresses automatic financial alert generation by combining three complementary signals: named entity recognition (NER), sentiment analysis (SA), and image captioning. Financial news often contains information that must be interpreted quickly in order to detect risks, opportunities, or relevant market events, and this project attempts to identify relevant signals for generating actionable financial alerts.

Starting from the FiNER-139 dataset, we first reduced the original label space to a smaller set of financially relevant entities and removed non-informative samples in order to obtain a cleaner and more task-oriented corpus. Since the dataset did not include sentiment labels, we reconstructed each text from its tokens, used FinBERT to generate pseudo-labels, and then trained our own sentiment classifier. In parallel, we explored visual context by generating short captions for a selected subset of news-related images using LLaVA-Phi3 through Ollama. These signals were then integrated into a prompt-based alert generation pipeline using an LLM. We compared several configurations, including isolated and combined uses of NER, SA, and image captioning, in order to understand which combination of components could produce higher-quality alerts and what differences the generated alerts might exhibit. Overall, the results show that the best performance comes from combining complementary signals rather than relying on any single component alone.

2. Prior related work

Automatic analysis of financial news requires combining different types of models, since a useful alert does not only depend on detecting relevant information in the text, but also on interpreting its tone and, in some cases, on adding visual context.

For NER, one of the best-known classical architectures is BiLSTM-CRF, which has been widely used in sequence labeling tasks because it can use the context of each word and produce coherent label sequences. Alongside these approaches, models based on BERT and other Transformers have become a more recent reference for NER, including applications in specialized domains such as finance, where the goal is not to detect general entities, but more specific concepts such as debt, revenue, or interest rates.

In sentiment analysis, the financial domain has an additional difficulty, because many texts look neutral in form even though they describe clearly positive or negative situations from an economic point of view. For this reason, Transformer models adapted to financial language have become important, especially FinBERT, which is widely used for polarity classification in financial text. This specialization is especially relevant in our work, since we used FinBERT in the preprocessing stage to generate sentiment pseudo-labels for the original corpus.

For image captioning, pretrained models can convert an image linked to a news item into a short textual description, making it easier to add visual information

to the pipeline without training a full multimodal system from scratch. Models such as BLIP, BLIP-2, and the LLaVA family follow this idea. In our case, this line of work supports the use of llava-phi3, which produced more precise and useful captions than a simpler pretrained baseline in our initial experiments.

For alert generation, recent NLP systems often rely on instruction-tuned generative models, which can transform structured inputs such as entities, sentiment, and captions into short and actionable summaries.

3. Models

3.1 Named Entity Recognition

The NER model uses a simple BiLSTM to assign one BIO tag to each token from a fixed vocabulary. This is a suitable choice for the project because it is lightweight, easy to train, and able to capture both left and right context, which is useful for identifying financial entities. However, its simplicity can reduce robustness in some cases: basic tokenisation means the text is split into tokens using simple rules, while unknown-word handling means that words not seen during training are mapped to a generic UNK token, causing some specific information to be lost. Even so, the overall expected performance is still good for the scope of this project.

3.2 Sentiment Analysis

The SA module uses a BiLSTM to classify each text as positive, neutral, or negative. The model takes the tokenized text of each sample, represented as a sequence of indices from a custom vocabulary, and processes it through an embedding layer followed by a BiLSTM. A final linear layer then produces the sentiment class.

This model is trained on the enriched corpus created in the preprocessing stage. This architecture is a good fit for the project because it keeps the model relatively simple while still capturing contextual information that is important in financial language. In this domain, sentiment often depends on the full sentence context rather than on isolated words, so a BiLSTM provides a reasonable balance between simplicity and contextual understanding.

3.3 Image Captioning

We initially tested a standard pretrained image captioning model, but its outputs were often too weak and inaccurate for our images, so we replaced it with LLaVA-Phi3 served locally through Ollama. Ollama allowed us to run a stronger multimodal vision-language model and query it directly from Python by sending each image together with a prompt asking for a single caption. This change was made to obtain more reliable and semantically richer descriptions, and it led to clearly better captions than the initial baseline. The final output of this stage is a JSON file containing the generated caption for each image, ready to be used in the rest of the pipeline.

3.4 Alert Generation Model

The alerts module relies on a language generator responsible for producing the final financial alert in natural language, based on the previously processed information. The system originally supported two alternatives for this generation phase. The first consists of using prompting with Ollama, sending the generative model a structured prompt with the relevant information extracted beforehand. The second option consists of performing fine-tuning with Hugging Face, training a generative model on examples specifically prepared for the alert generation task. In this way, the system allowed both an approach based on direct inference with instructed models and another based on supervised adaptation. Finally, it was decided to work with Ollama and the qwen2.5:3b model, since it made it possible to adjust the system’s behavior by modifying only the prompt, without the need to retrain the model in each experiment.

4. Data

4.1 Original dataset

We started from the FiNER-139 dataset, which contains 139 fine-grained financial NER tags and a high proportion of non-entity (“O”) tokens, making it unsuitable for our final objective. To address this, we performed an initial preprocessing step consisting of: (i) selecting a subset of 10 financially meaningful entities aligned with sentiment interpretation (e.g., revenues, losses, debt), (ii) filtering the original annotations by keeping only these entities while reassigning all others to “O” and preserving the BIO format, and (iii) removing samples containing only “O” tags to reduce class imbalance and increase informative signal. Finally, we generated reduced .jsonl files and defined the corresponding vocabulary and label set, resulting in a cleaner and more task-aligned dataset for subsequent NER and sentiment analysis modeling. In addition, we built the NER vocabularies required for the task, including both the token vocabulary and the reduced label set

4. 2 Preprocessing NER

The preprocessing for the NER model consisted of converting tokens and BIO tags into numerical indices, padding sequences to create uniform batches, and mapping unseen words to the special <UNK> token.

4.3 Sentiment label construction

Since the original corpus was prepared for NER, each sample only contained tokens and ner_tags, but no sentiment labels. To add the SA task, we first rebuilt the full text of each sample from its tokens and then used FinBERT to generate sentiment pseudo-labels. In this way, each example was enriched with a sentiment field, with one of the three project classes (positive, neutral, negative), and a sentiment_score field, which stores the confidence of the prediction.

During this process, we found a strong class imbalance: most examples were labeled as neutral, while positive and negative were much less frequent. To reduce this problem, we lowered the decision threshold for the minority classes and set a value of 0.3 to assign a positive or negative label. Our goal was to

obtain more useful pseudo-labels for later training. The result of this stage was an enriched corpus that could later be used to train the SA module and integrate sentiment information into the full alert generation pipeline.

4.4 Image-caption subset

Since the original dataset does not contain visual data, we are not working with images for all samples. Instead, we selected a small subset of representative examples, generated an associated image for each one using AI, and then applied an image captioning model to convert each image into text, as explained in the image caption model before.

4. 5 Alert preprocessing

The data preprocessing of the alert’s module starts from the outputs generated by the NER and SA models, which act as input for the subsequent generation phase. First, the output of the NER model, initially expressed in BIO format at token level, is transformed into complete entities by grouping consecutive labels that belong to the same tag. This makes it possible to work with complete financial expressions rather than isolated fragments of the text. In parallel, the output of the SA model is normalized to obtain a consistent label, limited to the positive, negative, or neutral classes, since it previously also returned a dictionary with the probability of each class.

Based on this preprocessed information, the prompts that will serve as input to the alert generator are built. These prompts integrate in a structured way the relevant financial entities, the detected sentiment and, optionally, may also incorporate the news text. It was necessary to carry out several prompt engineering tests to find the most suitable way of transforming the entities and sentiment into a useful alert. During these tests, it was observed that overly open prompts generated more natural alerts but were also more likely to add interpretations or information not explicitly present, whereas excessively rigid prompts improved formal uniformity, although they produced outputs that were too mechanical. Therefore, an intermediate solution was adopted, with instructions clear enough to control the structure, respect the detected entities and the associated sentiment, while still leaving some room for the model to write brief and fluent alerts.

5. Experiments

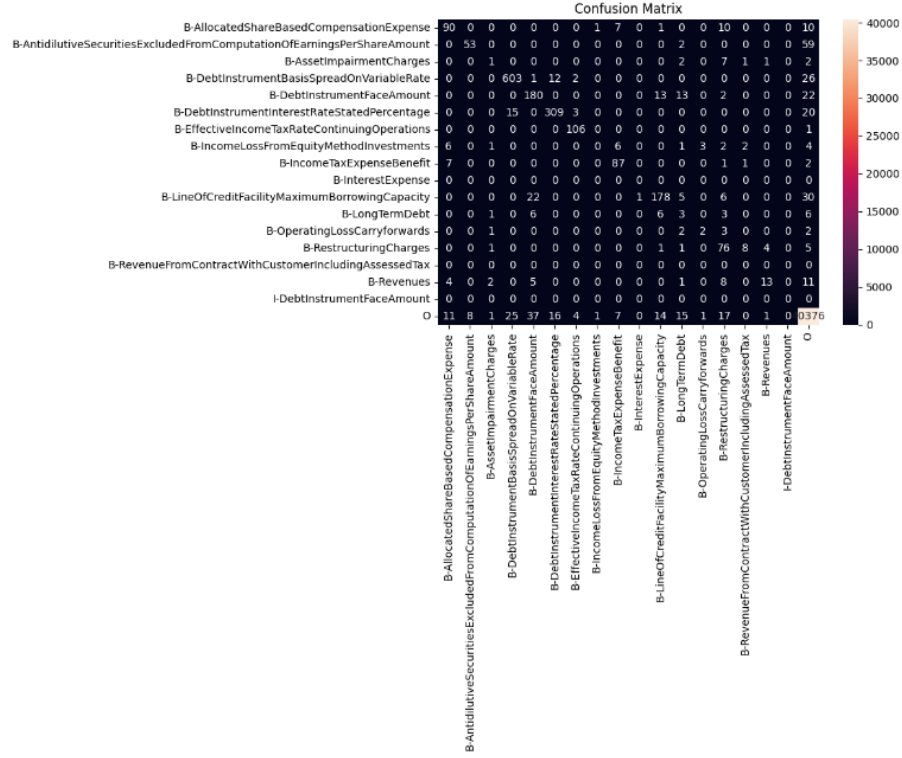


Figure 1: confusion matrix on the NER model

This confusion matrix for the NER model was obtained by counting how many times each predicted label matches or differs from the true label across the dataset. We chose this metric because it allows us to see how predictions are distributed across classes. In particular we could see most predictions lie on the diagonal, indicating good model performance. Although there is a strong imbalance with many “O” labels that can inflate accuracy, this is expected since there are usually more “O” tokens than entities, overall, the diagonal remains solid, indicating good performance.

NER loss

NER reaches 0.9866 test accuracy, although this value may be partly inflated by the large number of O labels, the model still shows strong real performance in financial entity detection.

SA loss

SA achieves 0.8460 test accuracy; it is slightly lower, and one possible factor is dataset imbalance (neutral: 397/500 vs. 76 negative and 27 positive), however it is still a very good result.

NER + SA loss

A multitask model was trained with a joint loss $L = L_{\text{NER}} + L_{\text{SA}}$, optimizing NER and SA simultaneously over a shared text representation. On validation, NER achieves very strong performance ($\text{ner_loss} = 0.0642$, $\text{ner_acc} = 0.9836$) and SA, while slightly lower, still shows good results ($\text{sa_loss} = 1.5471$, $\text{sa_acc} = 0.8520$). Training them together can be useful because both tasks support each other and this may result in better alerts in the next phases.

Combination of models:

MODEL	NER	SA	Image Caption- ing	JSON Name
Model 1	x			alert_predictions_ner_only.json
Model 2		x		alert_predictions_sa_only.json
Model 3			x	alert_predictions_caption_only.json
Model 4	x	x		alert_predictions_ner_sa.json
Model 5	x		x	alert_predictions_ner_caption.json
Model 6		x	x	alert_predictions_sa_caption.json
Model 7	x	x	x	alert_predictions_ner_sa_caption.json

The following analysis of the seven models is qualitative and based on a general review of the 20 generated prompts and our own interpretation of the outputs. Therefore, it should not be understood as an exact or fully objective evaluation, but rather as an indicative analysis that can help identify useful patterns and draw meaningful conclusions. Furthermore, some of the ideas extracted from each JSON file do not apply to each output in the JSON, but only to some specific cases; however, we consider them to reflect the main differences between the outputs of the different models.

Model 1: Ner

alert_predictions_ner_only.json captures the main financial event in many cases. It often preserves relevant financial entities, which makes the generated alerts informative and generally faithful to the original text. However, its precision is not fully consistent, because some outputs repeat numerical values, rely too heavily on extracted entities. In addition, several alerts appear truncated or incomplete, which reduces their reliability as final outputs.

Model 2: SA

alert_predictions_sa_only.json contains alerts generated only from sentiment. These alerts are usually clear and easy to understand, because the model captures the overall tone of the text and uses it to produce either a positive alert or a financial risk alert. This gives the output a natural style and a clear direction. In positive cases, it often treats higher borrowing capacity or financing extensions as favorable developments, while in negative cases it highlights restructuring costs, impairment losses, or tax increases. However, this setting has a clear

limitation: since it does not use NER, the alerts are less precise and are not tied to explicit financial entities. As a result, some outputs are more interpretative, and a few alerts become too long or end up truncated.

Model 3: Image Captioning

`alert_predictions_caption_only.json` appears to preserve concrete financial information more effectively in this sample, as it tends to retain amounts, credit increases, and maturity dates with greater precision. For example, when a company expands its financing, the alert usually reflects the higher credit limit and extended deadline directly, rather than replacing those details with more generic wording.

Model 4: NER + SA

`alert_predictions_ner_sa.json` combines NER and SA to produce alerts that are more detailed and more closely linked to the financial content of the text than the alerts generated with SA alone. In general, sentiment helps define whether the alert should sound positive or risky, while NER adds more specific information such as amounts, credit limits, interest rates, restructuring charges, or tax rates. This makes many alerts more informative and more precise. However, this setting has some limitations. The final quality depends on the entities extracted by NER, and in some cases these entities are repeated, incomplete, or unnatural, so the alert may rely too much on isolated numbers. In addition, the generator sometimes uses technical NER labels directly instead of turning them into natural language, and some outputs are still too long or cut off at the end. Overall, NER + SA improves over SA alone in many cases, but it still has some problems with fluency and robustness in the final generation.

Model 5: Ner + Image Captioning

`alert_predictions_ner_caption.json` generally has the potential to produce more context-aware financial alerts. Image captioning makes the output more useful than a purely generic summary, and it especially helps the model connect the extracted entities more naturally. However, some alerts still over-rely on extracted numbers and repeat financial values.

Model 6: Image Captioning + SA

`alert_predictions_sa_caption.json` is stronger at capturing the overall tone of the alert, especially in negative cases, because losses or restructuring events are more clearly framed as risk-related warnings. Although this reduction in precision cannot be treated as a general rule from only 20 examples, in this sample the inclusion of sentiment appears to make the model prioritise whether the news is positive or negative over the exact preservation of figures and details, leading to outputs that are sometimes shorter, vaguer, or incomplete; therefore, if exact numerical values are needed, the original message should still be checked directly.

Model 7: Ner + SA + Image Captioning

`alert_predictions_ner_sa_caption.json` is the most complete configuratio. In

this 20-example subset, the output seems to achieve a better balance between preserving key financial facts and reflecting the overall tone of the event. It has the potential to generate alerts that are both more informative and more context-aware, especially when numerical entities need to be retained while the alert is also framed with SA. However, this advantage should be interpreted cautiously, because the caption-based component is evaluated on only 20 examples, so it is not enough to claim a fully general improvement.

6. Analysis and conclusion

The comparison across the seven settings shows that the best results come not from adding components blindly, but from combining signals with complementary roles. Among the individual models, NER is the strongest for factual precision, because it anchors generation to explicit financial entities such as debt amounts, rates, taxes, or restructuring charges; however, by itself it can sound mechanical, over-focused on extracted spans, and sometimes unstable when entity extraction is noisy. SA alone is useful for controlling the tone of the alert and for rapid monitoring, since it helps the model quickly frame an event as positive, negative, or neutral, but it is weaker when the goal is to produce a compact and precise financial summary, because global sentiment tends to encourage interpretation rather than exact preservation of figures. Image captioning alone is not a sufficiently strong financial signal, but it can add complementary context when the visual content is aligned with the event. For this reason, the strongest text-only combination is NER + SA, which balances factual grounding and polarity framing better than either component alone, while the most complete configuration is NER + SA + image captioning, which achieves the best overall balance in the 20-example caption subset. This result makes sense from a deep learning perspective: NER provides local token-level grounding, SA provides sentence-level semantic orientation, and captions contribute orthogonal multi-modal context. Still, the gain from captioning should be interpreted cautiously because it was tested on a small subset, and all conclusions remain dependent on the specific Ollama-based generator and prompting setup used in this project.